

INVESTIGATIVE VERIFICATION LAB

VeriFact

Evidence-grounded misinformation detection

Retrieval-first verification of claims and links, where every verdict traces back to the sources that produced it.

[Presenter name] · [Course] · [Date]



Fluent is not the same as true

01

Chatbots answer from memory

Ask a language model whether something is true and it replies with confidence, but gives no source you can check.

02

Human fact-checks do not scale

Professional reviews are careful but slow, and a claim can travel widely before one is published.

03

A bare score hides the work

A label like "87% fake" does not show what was read, what was ignored, or how much each source counted.

Models read evidence. Rules decide.

The language model never answers from memory. It only reads passages we retrieved, and a deterministic, testable rule turns that evidence into a verdict.

Four layers, one audit trail



THE WORKER CALLS



Every progress event is stored in the database and replayed over SSE, so a page refresh or worker restart loses nothing.

From input to verdict in nine steps

01 Ingest

SSRF-safe fetch of text or a URL

02 Extract claims

Atomic, checkworthy, tied to a source span

03 Plan queries

Confirming and disconfirming searches

04 Retrieve

Search, news, Wikipedia, PubMed, Crossref

05 Read and rank

Best passage per page, BM25 then cross-encoder

06 Deduplicate

A syndicated story counts only once

07 Judge stance

Supports, refutes or neutral, with a verbatim quote

08 Weigh sources

Transparent rule-based credibility prior

09 Aggregate

Rule-based verdict, or Unverifiable

Guardrails enforced in code

No answers from memory

The model only reads passages we retrieved.

Quotes must be real

A stance is discarded unless its quote appears verbatim in the passage.

No forced verdicts

Thin or conflicting evidence ends as Unverifiable or Disputed.

Stricter when it matters

Health, election and violence claims need three independent domains.

Honest confidence

Computed from agreement and source weight, and labelled uncalibrated.

Limits travel with results

Every result lists what could not be verified.

How a verdict is computed

WEIGHT OF EACH SOURCE

$\text{credibility} \times (0.5 + 0.5 \times \text{relevance})$

SHARE OF SUPPORT

$p = \text{support} / (\text{support} + \text{refute})$

- At least 2 independent domains (3 for sensitive topics)
- Total weight of at least 0.9
- Strong disagreement gives Disputed, with no confidence number

Support share p

Verdict

0.85 or more

True

0.65 to 0.85

Mostly True

0.35 to 0.65

Mixed

0.15 to 0.35

Mostly False

0.15 or less

False

What is built and tested

39

automated tests passing across
three codebases

2

neural models checked on real
downloaded weights

5

keyless evidence sources checked
against live services

3

LLM providers behind one
interface, Groq free by default

Backend

Nine-stage pipeline, source
adapters, SSRF-safe fetcher,
schema-validated LLM layer, SSE
progress.

ML service

Cross-encoder reranker, DeBERTa
NLI, and error-level and noise image
forensics.

Eval harness

Precision, recall, F1, AUROC and
ECE, plus a runner that drives the
real pipeline.

The reasoning is on screen

Live investigation

Real pipeline events, not timers.

Evidence Board

Stance by shape, size by credibility, exact excerpt on hover.

Verdict card

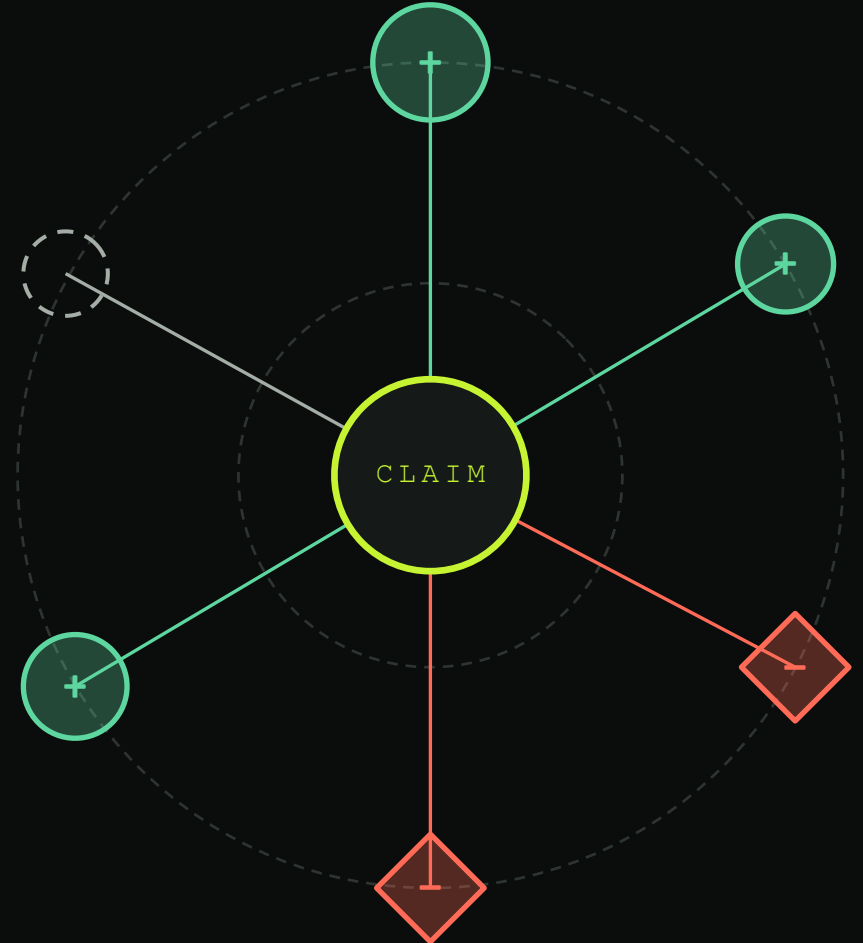
Evidence strength, plus what we could not verify.

Show the reasoning

Queries run and sources discarded, with reasons.

Model toggle

Free Groq, or paid Anthropic and OpenAI.



Choices and their costs

Decision	Why	Cost
Rules decide the verdict	Inspectable and unit-testable	Less flexible than an LLM judge
Free-first stack: Groq and keyless sources	Reproducible with no budget	Rate limits; unofficial search endpoint
Separate ML service	Heavy dependencies stay out of the API	An extra container; BM25 fallback
Verbatim-quote check	Blocks invented justifications	Cannot catch a real quote misread

One bug found along the way: the standard BM25 variant gave zero scores on small candidate pools, so ranking was silently dead until a test exposed it.

What we can and cannot claim

Ready

- Precision, recall, macro-F1 and AUROC
- Calibration: ECE and reliability bins
- Runner that sends labelled claims through the real pipeline
- "No verdict" counted as its own class

Not yet done

- A benchmark run (LIAR first, then FEVER)
- Calibrated confidence on a held-out split
- Media benchmarks for images, video and audio
- Threshold tuning against real results

No accuracy numbers are quoted here, because none have been measured yet.

Known weaknesses

01 Confidence is a heuristic, not a probability.

02 Credibility uses rule-based priors, not licensed ratings.

03 Stance is judged by an LLM on short passages; free models are weaker.

04 Free web search is unofficial and can fail or skew.

05 Images, video and audio are not yet in the live flow.

06 Live-web evaluation drifts over time and can leak answers.

Seven milestones, honestly staged

#	Milestone	Status
1	Text and URL pipeline with evidence UI	● Built
2	Image pipeline	◐ Partial: ELA and noise forensics
3	Audio and video	○ Planned
4	Fusion and calibration	○ Planned
5	Evaluation runs	◐ Tooling built, runs pending
6	Interface polish	◐ Partial
7	Hardening: rate limits, retention, auth	○ Planned

L I V E D E M O

Claim in, evidence out.

Watch the pipeline run, open the evidence, and read the reasoning behind the verdict.

Questions?

[Repository link or contact]